Transcriptomic analysis of EV miRNA cargo derived from allergen-exposed airway epithelial cells using open-access software

Christopher Janschke

20805916

MSc in Bioinformatics
School of Life Sciences
Queen’s Medical Centre

Supervisor: Professor Lucy Fairclough, Dr Georgina Hopkins, Dr Hannah Jackson

Running title: Transcriptomic Profiling of Extracellular Vesicles Released by Human Bronchial Epithelial Cells

This dissertation is submitted in partial fulfilment of the project requirements for the M.Sc.:

Bioinformatics
July 2026

Abstract:	226 words	Number of tables:	0
Main text:	4794 words	Number of figures:	6

GitHub repository: https://github.com/CJ770411/L4137_HBEC_EV_Transcriptomics

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Abstract

Respiratory diseases constitute a significant health burden worldwide. As well as genetic factors, environmental factors, such as exposure to inhaled allergens, can drive the prevalence of these diseases. Allergen exposure can alter the miRNA cargo of extracellular vesicles (EVs) derived from airway epithelial [cells]. Through the post-transcriptional repression of target genes, altered miRNA cargo has the potential to disrupt epithelial barrier homeostasis, triggering inflammation and potentially driving allergic sensitisation. Allergen-exposed EV miRNA cargo and affected downstream pathways can be hypothesised using transcriptomic analysis, though this is often outsourced and performed through commercial platforms, which can be costly. Here we show that biologically plausible transcriptomic analysis can be performed using exclusively open-access software. We found distinct EV miRNA signatures in primary human bronchial epithelial cells exposed to high doses (24 μg/cm^2) of nebulised protease allergen, which may influence pathways associated with immune and inflammatory signalling as well as epithelial barrier regulation. Whereas transcriptomic analysis for Calu-3 cells was precluded due to insufficient EV material. These results demonstrate that the transcriptomic workflow can detect credible biological responses associated with allergen exposure when provided appropriate raw experimental data. This study provides experimental design recommendations for future allergen studies to optimise transcriptomic analysis. Furthermore, we provide a rationale for performing differential expression analysis of EV-derived miRNA cargo in-house whereas robust pathway enrichment analysis may be enhanced by using commercial platforms.

Introduction

Allergy-induced respiratory diseases constitute a significant health burden, with an estimated 260 million cases of asthma existing worldwide in 2021 (Muo et al., 2025). Environmental and genetic factors during development drive the prevalence of chronic airway inflammation diseases such as asthma (Kaur and Chupp, 2019). The airway epithelial barrier constitutes the primary line of defence against inhaled allergens in healthy individuals and is crucial for maintaining homeostasis (Laulajainen-Hongisto et al., 2020). It exists predominantly as a single-layer comprising various types of cells (e.g., epithelial, basal, goblet) which interact with a variety of immune cells to regulate inflammatory responses (Jesenak et al., 2026). Disruptions to barrier integrity is indicated in many respiratory diseases (Carlier et al., 2021). Extracellular vesicles (EVs) have been demonstrated to be involved in the pathogenesis of respiratory diseases and airway remodelling (Mohammadipoor et al., 2023; Sangaphunchai et al., 2020). This suggests EVs may have the potential to disrupt the epithelial barrier and propagate airway inflammation.

EVs are small lipid bilayer-enclosed particles, released almost ubiquitously among human cells, responsible for intercellular communication via transfer of their bioactive cargo to recipient cells (Chen et al., 2024; Kadota et al., 2021; Fujita et al., 2024). This cargo consists of various biomolecules including microRNAs (miRNAs), which are small non-coding RNAs approximately 18-30 nucleotides (nt) that exercise gene silencing through post-transcriptional degradation of messenger RNAs (mRNAs) (Levänen et al., 2014). Lung epithelial cells readily generate EVs during lung inflammation, which can be commonly triggered through allergen exposure. (Sangaphunchai et al., 2020; Holtzman and Lee, 2020). EV miRNA cargo has been shown to change upon exposure to allergens and between healthy individuals and respiratory disease sufferers (Bartel et al., 2019; Levänen et al., 2013; Gon et al., 2017). Common allergens known to trigger inflammatory responses include house dust mite (HDM), dog allergens and protease enzyme allergens, found in many household cleaning products (Hopkins et al., 2026; Browne et al., 2026). Transcriptomics have emerged as a valuable tool to investigate miRNA cargo and subsequently investigate the biological role of differentially expressed miRNAs.

Advances in dose-delivery technologies have improved allergen exposure studies by enabling aerosolised delivery, which more closely reflects *in vivo* airway exposure conditions compared with traditional liquid-based delivery systems. Previous studies have demonstrated that liquid-based allergen exposure can alter epithelial barrier integrity and alter the transcriptome in primary human bronchial epithelial cells (HBECs) (Mallek et al., 2024). Two previous studies have adopted nebuliser technology to investigate the miRNA cargo of allergen-exposed airway epithelial cell-derived EVs and the biological relevance of pathways associated with differentially expressed miRNAs (Hopkins et al., 2026; Browne et al., 2026).

In both studies, the transcriptomics analysis was performed using commercial services. The disadvantages of this are that it can be costly and in the case of closed-source software, locks reproducibility behind a paywall.

This study aimed to create a transcriptomics workflow to re-analyse small RNA sequencing (RNA-seq) data generated in previous allergen studies Hopkins et al., 2026 and Browne et al., 2026. Whereas the previous analyses employed commercial services and platforms to conduct transcriptomics, this study uses exclusively open-source and open-access software. The workflow integrates sequencing data quality control, differential expression analysis and evaluates two approaches to pathway enrichment analysis, with a particular focus on identifying immune and inflammatory pathways relevant to airway biology. At the time of this study, pathway analysis had not been performed on the Browne et al. 2026 dataset; accordingly, this work represents a novel analysis. Collectively, this work establishes a reproducible, open-access workflow for the analysis of EV-derived miRNA sequencing data and highlights methodological considerations for improving the experimental design and transcriptomic analysis of future allergen exposure studies.

Methods

Workflow

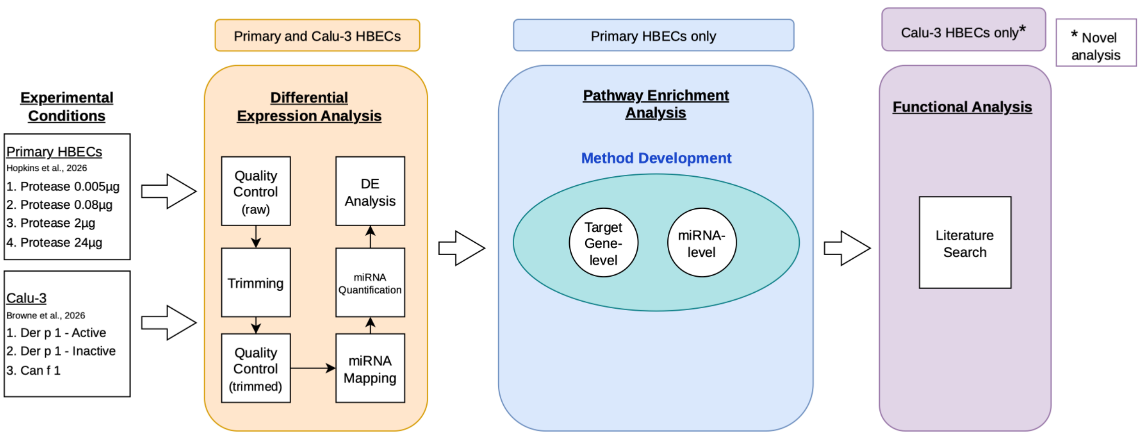

Figure 1: Workflow outline. Previously acquired raw sequencing data for primary human bronchial epithelial cells (HBECs) exposed to 0.005, 0.08, 2 and 24 µg/cm² protease allergen was obtained from Hopkins et al., 2026, and for Calu-3 lung adenocarcinoma cells exposed to Der p 1 active and Der p 1 inactive (house dust mite allergens) and Can f 1 (dog allergen) obtained from Browne et al., 2026. The differential expression phase (orange) was performed using primary HBECs and Calu-3 datasets. First, quality control was performed on raw sequencing data, then reads were trimmed and quality control was repeated. miRNAs were mapped to known mature miRNAs and quantified. Differential expression analysis was then performed by comparing each allergen exposure condition in primary HBECs and Calu-3 to a corresponding untreated group. The next phase (blue) focused on developing an optimal method for pathway enrichment analysis for differentially expressed (DE) miRNAs; only 24 µg/cm² protease-treated primary HBECs were used for this phase. The final phase (purple) used literature to infer the biological function of DE miRNAs derived from the Calu-3 dataset. Figure generated using draw.io.

These methods describe an independent analysis pipeline established prior to methods or results from any previously performed bioinformatics analysis being disclosed.

Initially, only the primary HBEC dataset was made available, and these data were analysed to completion of pathway enrichment analysis method development before the Calu-3 dataset was made available.

Study data

This study uses raw next-generation sequencing (NGS) data acquired from small RNA-Seq using the miND® analysis pipeline (Khamina et al., 2022). RNA was acquired from airway epithelium-derived EVs following cellular exposure to nebulised allergens, according to the methodology set out in (Hopkins et al., 2026; Browne et al., 2026). Both papers were submitted for review at the time of this study but not yet published.

Hopkins et al., 2026 exposed primary HBECs to 0.005, 0.08, 2 and 24 μg/cm^2 protease enzyme allergen. Browne et al., 2026 exposed Calu-3 cells (lung adenocarcinoma cell line) to nebulised Can f 1 (dog allergen), active Der p 1 and inactive Der p 1 (HDM allergens). All conditions were executed in triplicate.

Preprocessing

Raw sequencing reads were trimmed using Cutadapt (v5.2), targeting the Realseq3P adapter (TGGAATTCTC). Only reads measuring 18-30 nt and quality >30 (Phred) were retained; the first base of each read was unconditionally removed and reads without an adapter detected were discarded. Quality control was performed using FastQC (v0.12.1) and miRTrace (v1.0.1) and reported with MultiQC (v1.34).

Mapping and quantification

Reads were collapsed using the miRDeep2 (v2.0.1.3) mapper.pl script then mapped to mature and hairpin references (miRBase v22.1) and quantified using quantifier.pl, allowing for 1 mismatch.

Differential expression analysis

DE analysis was performed using edgeR (v4.8.2). Lowly expressed miRNAs were excluded using the default filtering provided by the package. Normalisation was performed using the trimmed mean of M-values normalisation and dispersion estimated using the negative binomial dispersion by weighted likelihood empirical Bayes. DE miRNAs were obtained from pairwise contrasts (treated versus untreated) using the quasi-likelihood F-test. False discovery rate (FDR) was controlled using the Benjamini-Hochberg (BH) multiple testing correction method. DE miRNAs were classified as significant based on a cut-off of log 1-fold-change and FDR < 0.05. An exploratory cut-off was set at FDR < 0.2 if few DE miRNAs were initially detected.

Pathway enrichment analysis (method development)

Multiple automated pathway enrichment analysis techniques were explored as part of a method development phase intended to determine the optimal technique for use on the novel dataset acquired from Calu-3 cells (Supplementary Table 1). DE miRNAs identified in the 24 μg/cm² protease-treated primary HBEC group were used to test one free-of-charge target gene-level and miRNA-level enrichment tool. Significantly enriched pathways were then compared to pathway results obtained from one monetised pathway analysis service provider.

Target gene-level enrichment

Experimentally validated target genes of significantly DE miRNAs were identified using multiMiR (v1.34.0) R package by querying miRTarBase (v9). Pathway enrichment analysis of the target genes was performed using the enrichPathway() function from the ReactomePA (v1.56.0) R package, which implements a hypergeometric over-representation test to identify Reactome pathways enriched among the target genes. The default background universe was used. The BH correction was applied to control the FDR. Only pathways containing 10-500 genes were tested; pathways with an adjusted P value < 0.02 were considered significantly enriched.

miRNA-level enrichment

miRNA lists were generated in RStudio (v4.6.0) and submitted to miEAA (v2.1, accessed 10th July 2026) to identify enriched Reactome (miRPathDB) pathways. For over-representation analysis (ORA), the test set comprised significantly upregulated or downregulated DE miRNAs, with all miRNAs detected in the experiment used as the background universe. To identify pathways enriched by coordinated, but not necessarily DE, miRNAs, miRNA-set enrichment analysis (miRNA-SEA) was performed using all detected miRNAs ranked by log fold change (logFC). For both analyses, adjusted p-values were calculated using BH correction for FDR, and pathways with adjusted P value < 0.05 were considered significantly enriched.

Literature search

Pathway enrichment analysis was not applied to Calu-3 cells as few DE miRNAs were identified, therefore statistical power was limited. Instead, a literature search was conducted to investigate whether the DE miRNAs identified in Calu-3 cells had previously been associated with biological processes relevant to airway immune signalling and allergic sensitisation. For each DE miRNA, the miRNA symbol was used as the query term in PubMed (National Library of Medicine, accessed 12th July 2026) and combined with each of the following keywords using the Boolean operator “AND”: "airway", "immune", "allergic", "inflammation", "epithelial" and “extracellular vesicle”.

Statistics and data analyses

DE miRNAs with FDR < 0.05 were considered significant; FDR < 0.2 was applied as an exploratory significance threshold. Pathways identified through target-gene level enrichment were considered significantly enriched based on adjusted P value < 0.02 or through miRNA-level enrichment based on adjusted P value < 0.05.

Principal components analysis (PCA) was performed to assess variability in miRNA expression between cell-type specific allergen exposure conditions, and replicates. Principal components (PCs) were calculated using the prcomp() function in RStudio using log2 CPM-normalised counts with a nominal value of 2 added to prevent infinite values. Heatmaps were generated using log2 CPM-normalised counts. All figures were generated in RStudio (v4.6.0) using ggplot2 (v4.0.3), unless stated otherwise.

Statistical analyses were performed using R (v4.5.3). A full list of packages used can be found on the GitHub associated with this study along with references for each software tool.

Previous work

DE analysis was previously performed for all allergen exposures in primary HBECs and Calu-3 cells by TAmiRNA using the miND® pipeline (Khamina et al., 2022). Pathway analysis was performed using DE miRNAs in primary HBECs following exposure to dose 24 μg/cm^2

protease allergen by Prof. Victoria James using QIAGEN Ingenuity Pathway Analysis (IPA) (https://digitalinsights.qiagen.com/products-overview/discovery-insights-portfolio/analysis-and-visualization/qiagen-ipa/). Previous analysis and results are detailed in Hopkins et al., 2026 and Browne et al., 2026. Pathway analysis of DE miRNAs from Calu-3 cells in this study was novel.

Code availability

All code I developed for this study is available at https://github.com/CJ770411/L4137_HBEC_EV_Transcriptomics. I generated a custom-made template bash script (template.sh) to standardise script structure, perform log and error handling, and accelerate creation of new scripts. Access to project worklogs and all drafts can be requested from https://github.com/CJ770411/L4137_HBEC_EV_Transcriptomics-Project.

Results

Review of quality-control (QC) metrics to identify samples with unique QC profiles

Quality control metrics generated by FastQC and miRTrace using trimmed reads were analysed for all primary HBEC and Calu-3 samples to identify samples with distinct profiles (Supplementary Figure 1,2). The three 24 μg/cm^2 protease-treated primary HBEC replicates displayed distinct guanine-cytosine (GC) content, read-length, and sequence duplication profiles compared with the remaining primary HBEC samples and Calu-3 samples (Figure 2).

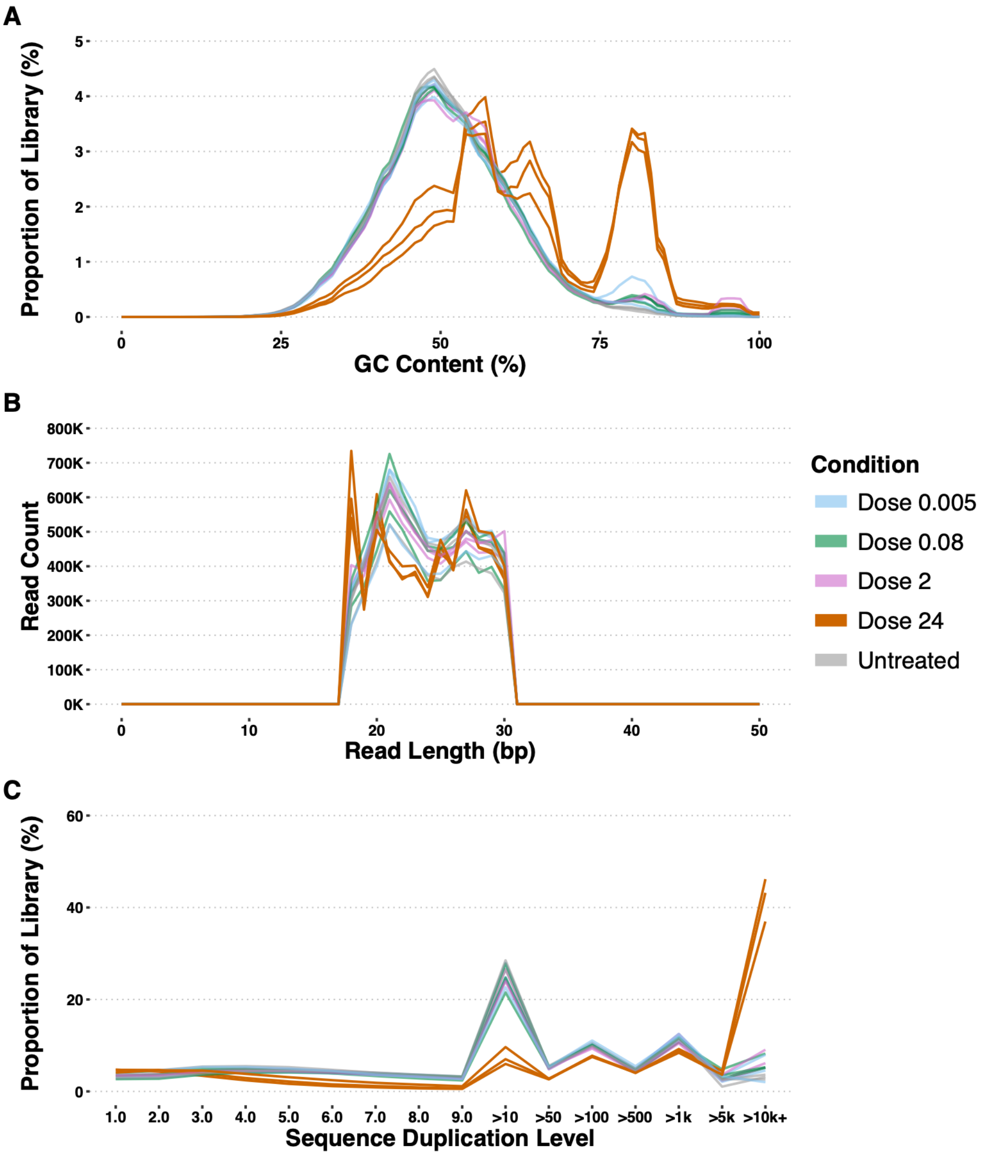

Figure 2: Primary HBEC samples sequencing library quality control. FASTQ files for all trimmed primary human bronchial epithelial cell (HBEC) samples (N=15) exposed to protease allergen were assessed for quality using FastQC (A, C) and miRTrace (B) then compiled into a single report using MultiQC. MultiQC plot data was then loaded into RStudio to generate the plots in this figure. (A) Read GC content distribution with per-read GC content shown as a percentage on the x-axis and the proportion of the whole library at a given GC content shown as a percentage on the y-axis. (B) Read length distribution with per-read length in base pairs on the x-axis and the number of reads of a given length shown on y-axis. (C) Read duplication levels calculated from the first 100,000 reads in each FASTQ file; FASTQC documentation states this should give a representative view of the whole library. The category of per-read sequence duplication is shown on the x-axis (higher duplication levels are shown towards the right of the x-axis) and the proportion of the whole library at a given duplication rate as a percentage on the y-axis. All plots are coloured by treatment group where dose is in μg/cm².

The modal per-read GC content of the 24 μg/cm^2 replicates was 57%, 80%, and 61%, respectively. In contrast, all other samples exhibited an approximately normal GC-content distribution with a modal GC content of ~49%. GC content across all trimmed reads was higher in the 24 μg/cm^2 replicates, with a mean of 62.67% compared to 50.75% in the remaining primary HBEC samples (Figure 2A), and 49.75% in Calu-3 samples (Supplementary Figure 1,2).

Read-length distributions ranged from 18–30 bp. The modal read length was 21 bp in all samples except the 24 μg/cm^2 replicates. In these samples, the modal read length was 18 bp, with a mean of 623,413 18 bp reads, compared with 314,956 in the remaining primary HBEC samples (Figure 2B), and 280,651 in Calu-3 samples.

The 24 μg/cm^2 replicates exhibited higher levels of sequence duplication than the remaining samples, containing a mean of 42.1% of reads with duplication levels greater than 10,000-fold, compared with 5.3% in the remaining primary HBEC samples (Figure 2C), and 3.7% in Calu-3 samples

Assessment of miRNA representation in trimmed read sequencing libraries

RNA composition was analysed using miRTrace for all trimmed primary HBEC and Calu-3 samples. For primary HBEC samples, the mean number of reads assigned to miRNAs by miRTrace was 44,681, representing 0.75% of total trimmed reads. For Calu-3 samples, the mean number of reads assigned to miRNAs was 2,496, representing 0.045% of total trimmed reads, with a range of 594–4,379 reads (Figure 3A).

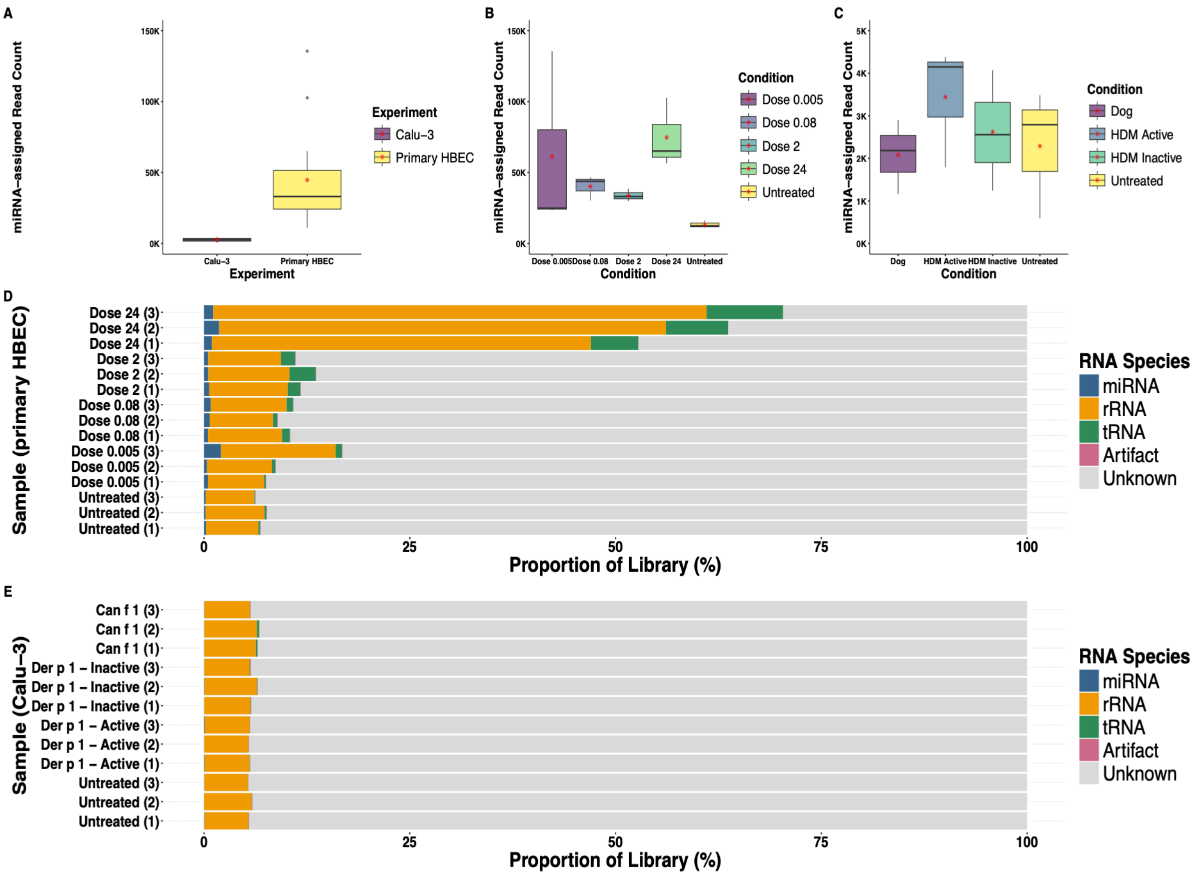

Figure 3: RNA composition analysis of primary HBEC and Calu-3 datasets. Sequencing reads for all trimmed primary human bronchial epithelial cell (HBEC) samples (N=15) and Calu-3 samples

(N=12) were categorised according to RNA species using miRTrace software then compiled into a single report using MultiQC. MultiQC plot data was then loaded into RStudio to generate the plots in this figure. (A) Boxplots showing the miRNA-assigned read count across all samples for the primary HBEC and Calu-3 experiments. The type of cells investigated in the experiment is shown on the x-axis and number of reads categorised as miRNA are shown on the y-axis; mean miRNA count is indicated with a red star. (B) Boxplots showing the miRNA-assigned read count across each exposure condition in the primary HBEC experiment, where there is n=3 samples per condition. Exposure condition is shown on the x-axis where dose is in µg/cm², the number of reads categorised as miRNA are shown on the y-axis; mean miRNA count is indicated with a red star. (C) Boxplots showing the miRNA-assigned read count across each exposure condition in the Calu-3 experiment, where there is n=3 samples per condition. Exposure condition is shown on the x-axis where “Dog” = Can f 1, “HDM Active” = Der p 1 (active) and “HDM Inactive” = Der p 1 (inactive); the number of reads categorised as miRNA are shown on the y-axis, mean miRNA count is indicated with a red star. (D, E) Sequencing library RNA composition of all primary HBECs (D) and Calu-3 (E) samples where each distinct RNA species is displayed as a proportion of the total library and indicated with a distinct colour. Samples are shown on the y-axis and are grouped by their experimental condition along with the replicate number which is presented in brackets. For primary HBEC samples, doses are in µg/cm². RNA species abbreviations: “miRNA” = micro-RNA, “rRNA” = ribosomal RNA, “tRNA” = transfer RNA.

To investigate the effect of allergen exposure on miRNA abundance, the number of reads assigned to miRNAs was calculated for each exposure condition. For primary HBECs, untreated samples had the lowest mean number of reads assigned to miRNAs (13,289; range: 11,098–16,332), whereas samples exposed to 24 µg/cm² had the highest mean (74,738) (Figure 3B). However, the number of reads assigned to miRNAs did not show a clear dose-dependent trend, with the 0.005 µg/cm² group having the second-highest mean, followed by the 0.08 and 2 µg/cm² groups, respectively. The elevated mean in the 0.005 µg/cm² group was influenced by a single replicate (replicate 3), which contained 135,524 reads assigned to miRNAs, the highest value observed among all samples. For Calu-3 samples, the Der p 1 active group had the highest mean number of reads assigned to miRNAs, whereas the Can f 1 group had the lowest (Figure 3C).

Analysis of other RNA species revealed that primary HBEC samples exposed to 24 µg/cm² contained a higher proportion of transfer RNA (tRNA) and ribosomal RNA (rRNA) than the other exposure groups. The mean proportion of rRNA in the 24 µg/cm² group was 53.4%, compared with a mean of 8.52% across the remaining primary HBEC samples (Figure 3D). In Calu-3 samples, only small proportions of reads were assigned to rRNA and tRNA (Figure 3E). These findings demonstrated that the 24 µg/cm² samples exhibited distinct RNA composition profiles compared with the other protease exposure conditions.

Overall, the lowest number of reads assigned to miRNAs in primary HBECs was observed in untreated samples. Furthermore, only a small proportion of trimmed reads from the Calu-3 sequencing libraries were assigned to miRNAs, which may limit the statistical power of downstream DE analysis.

Identification of differentially expressed EV-associated miRNAs

Prior to DE analysis, a principal components analysis was used to investigate the relationship between replicates and exposure conditions within primary HBEC samples and Calu-3 samples (Figure 4A,B). In primary HBEC samples, PCA showed separation between 24 µg/cm² protease-treated samples and untreated samples along PC1, which explained 25% of the variance. Some separation between replicates was observed in the untreated group (replicate 3) and the 0.005 µg/cm² group (replicate 2). Within the Calu-3 samples, Can f 1 (replicate 1) and untreated (replicate 2) samples showed separation from their corresponding replicates, whereas the remaining samples displayed clustering within their respective exposure groups. No samples were excluded based on PCA separation as each exposure group consisted of only three biological replicates and removal of samples would further reduce the statistical power of downstream DE analysis.

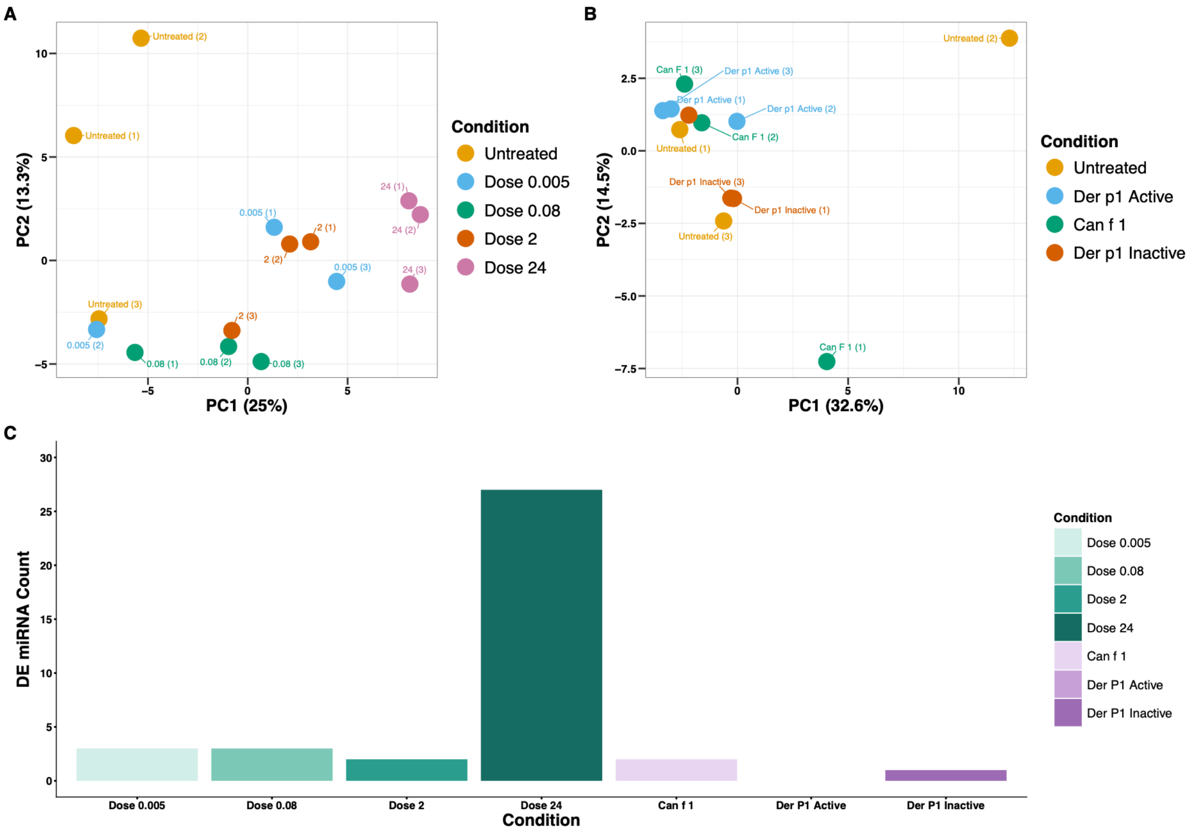

Figure 4: Principal components analysis and differential expression results. Sample-sample distances were calculated using the *prcomp()* function from the default *stats* package in RStudio based on log2 counts per million values for all samples retained after filtering in the *edgeR* differential expression workflow, with a prior count of 2 applied. Only the top two principal components (PCs) are shown; percentage variance explained is shown in brackets next to the PC on the x- and y-axes. Each dot represents an individual sample which is labelled with the condition and replicate numbers in brackets. Samples are coloured by allergen-exposure condition, where dose is in µg/cm². (A) Primary HBEC samples (N=15). (B) Calu-3 samples (N=12). Differential expression analysis was performed using the edgeR R package, using miRNAs detected across all exposure conditions in the primary human bronchial epithelial cell (HBEC) experiment (n=147) and Calu-3 experiment (n=58). (C) Differentially expression results from the primary HBEC and Calu-3 experiments. Allergen-exposure conditions are shown on the x-axis with primary HBEC conditions shown in green and Calu-3 conditions in purple; dose is in µg/cm². Bar chart generated in RStudio.

Reads mapped to known *Homo sapiens* miRNAs annotated in the miRBase database were analysed using edgeR to identify differentially expressed miRNAs between treated and untreated primary HBEC samples and Calu-3 samples. In primary HBECs, 147 of the 562 miRNAs detected across all protease exposure conditions were retained following removal of lowly expressed miRNAs. In the 24 μg/cm² exposure group, 27 miRNAs were significantly differentially expressed based on $|logFC| > 1$ and $FDR < 0.05$ (Figure 4C). The remaining protease exposure groups demonstrated little differential expression, with 2–3 differentially expressed miRNAs identified per condition (Figure 4C).

In Calu-3 cells, 58 out of the 263 detected miRNAs were retained following filtering. Given the low abundance of miRNA-assigned reads and the inconsistent replicate clustering observed in the PCA, differential expression analyses in Calu-3 cells were expected to have reduced statistical power. Two DE miRNAs were identified in the Can f 1 exposure condition using $|logFC| > 1$ and $FDR < 0.05$ (Figure 4C). Applying an exploratory significance threshold of $FDR < 0.2$ identified one additional DE miRNA was detected in the Der p 1 inactive condition; no DE miRNAs were detected in the Der p 1 active condition (Figure 4C). All DE miRNAs are detailed in Supplementary Table 2.

Overall, the greatest number of DE miRNAs was identified in the 24 μg/cm² protease-treated primary HBEC samples, whereas the remaining primary HBEC and all Calu-3 treatment conditions exhibited minimal differential expression.

Profiling of differentially expressed miRNA signatures

DE miRNAs were compared across exposure conditions within each cell type and between cell types to identify shared expression signatures. Within the primary HBEC experiment, miR-34b-3p was differentially expressed in all four protease exposure groups. miR-3960 was differentially expressed in the 0.08, 2 and 24 μg/cm² exposure groups, whereas miR-193b-3p and miR-148a-3p were differentially expressed in both the 0.005 and 24 μg/cm² exposure groups. Within the Calu-3 experiment, miR-574-5p was differentially expressed in the Can f 1 and Der p 1 inactive exposure groups. miR-125b-5p was the only miRNA that was differentially expressed across both cell types, being detected in the 24 μg/cm² protease-treated primary HBEC samples and the Can f 1-treated Calu-3 samples.

Further analysis focused on the 24 μg/cm² protease-treated primary HBEC group, which contained 27 DE miRNAs (Figure 5), whereas the remaining exposure groups contained few significantly DE miRNAs.

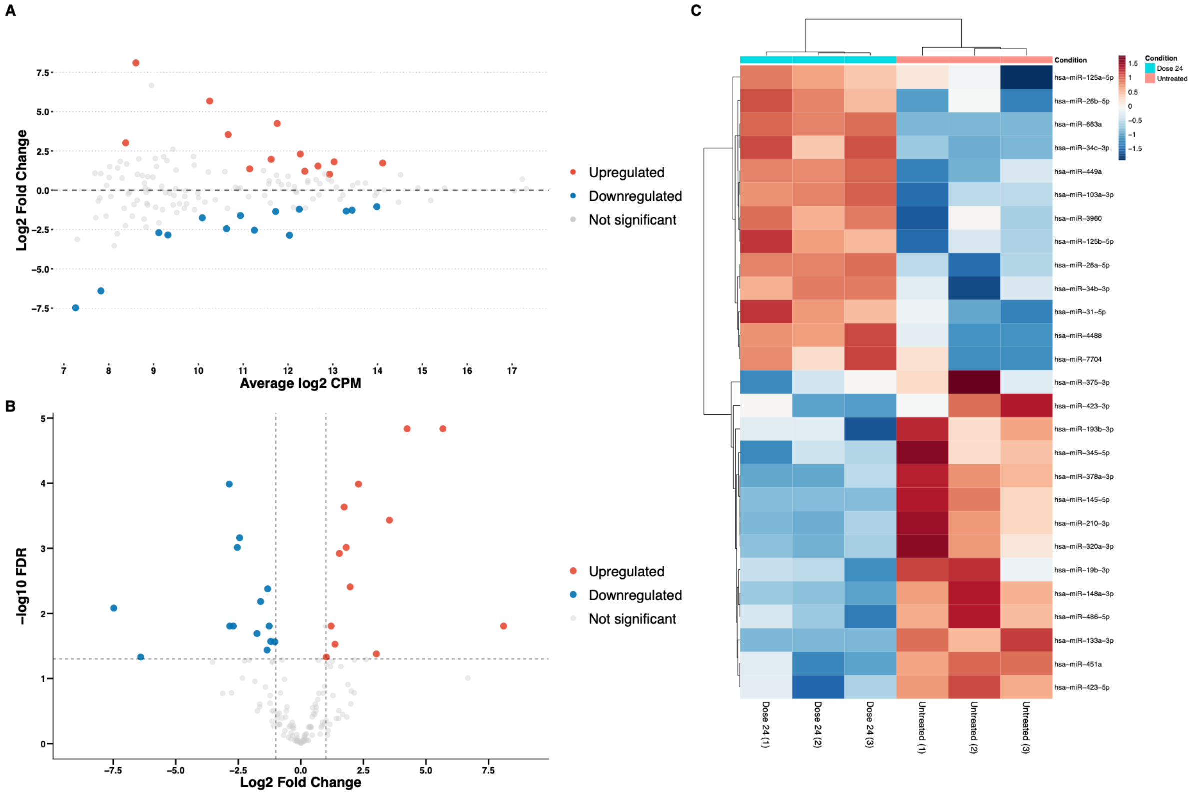

Figure 5: Analysis of differential expression in 24 µg/cm² protease-treated primary HBEC samples. Differential expression results generated from the *edgeR* R package for primary HBECs exposed to 24 µg/cm² protease allergen were visualised using plots generated in RStudio. (A, B) All miRNAs tested for differential expression (n=147) were plotted with each dot representing an individual miRNA coloured by significance. (A) The average log2 counts per million (CPM) on the x-axis and the log2 fold change in expression compared to untreated samples on the y-axis. (B) Log2 fold change in expression compared to untreated samples shown on the x-axis and significance (quasi-likelihood F test corrected for multiple testing using Benjamini-Hochberg) converted to -log10 scale on the y-axis. (C) Relative log2 CPM-normalised values for all significantly differentially expressed miRNAs shown for each replicate within the 24 µg/cm² protease-treated and untreated groups; replicate numbers are indicated in brackets next to the sample name on the x-axis, miRNA symbols are shown on the y-axis. Replicates and miRNAs are arranged according to closeness, as indicated with dendrograms. Dose is in µg/cm².

The distribution of differential expression in the 24 µg/cm² protease-treated primary HBEC samples was visualised using an MA plot, while statistical significance was displayed using a volcano plot (Figure 5A,B). Of the 27 differentially expressed miRNAs, 13 showed higher expression in the 24 µg/cm² samples, whereas 14 showed higher expression in untreated samples. Sixteen of the 27 DE miRNAs had an absolute logFC < 2.5 (Figure 5A). The smallest FDR observed was 1.46×10^{-5} (Figure 4B).

Relative normalised miRNA expression (CPM) in the 24 µg/cm² protease-treated and untreated samples was visualised using a heatmap to assess the concordance of expression among biological replicates (Figure 5C). The 24 µg/cm² samples generally exhibited consistent relative miRNA expression across replicates. In contrast, greater variability was observed among untreated replicates, with individual samples displaying distinct expression patterns for several miRNAs, including miR-375-3p (untreated replicate 2) and miR-7704 (untreated replicate 1).

Pathway enrichment analysis method development using primary HBECs

Before investigating the biological relevance of DE miRNAs identified in Calu-3 cells, the pathway enrichment workflow was developed and evaluated using the 27 DE miRNAs identified in the 24 μg/cm² protease-treated primary HBEC group (Figure 6).

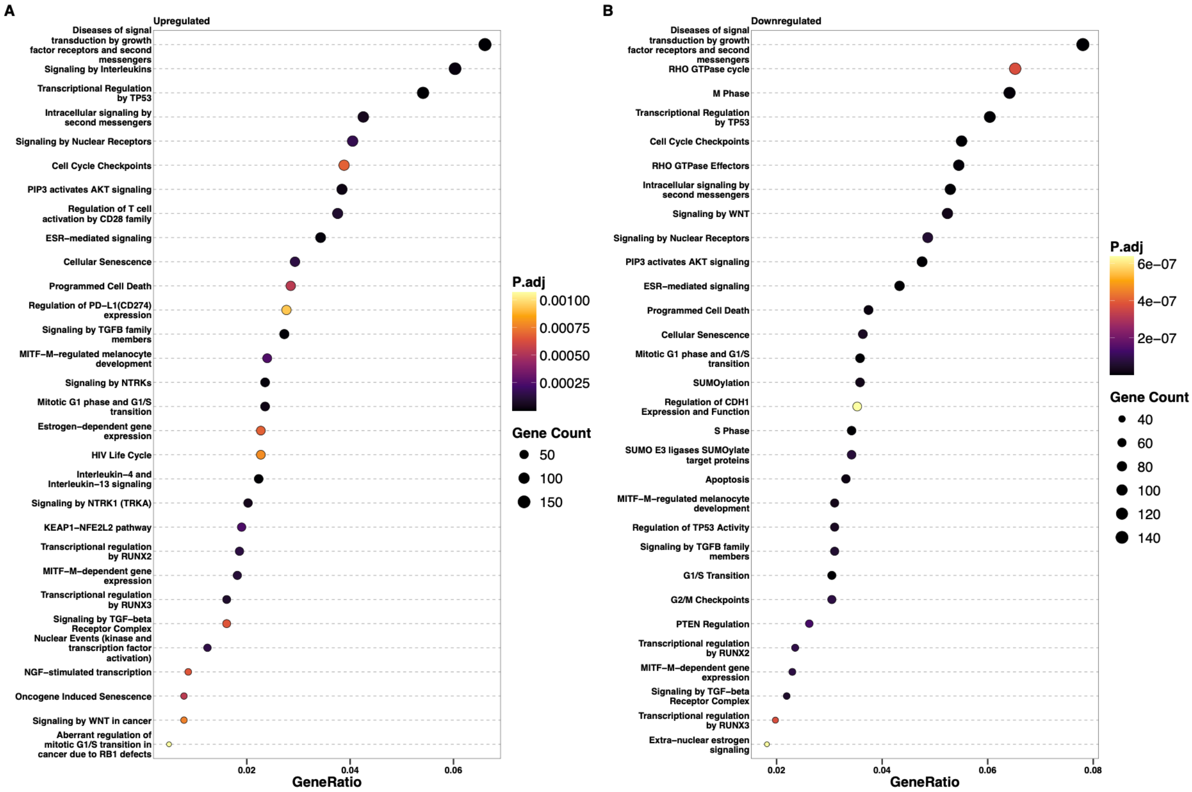

Figure 6: Target gene-level pathway enrichment results. Target genes for differentially expressed (DE) miRNAs (N=27) detected in primary human bronchial epithelial cells exposed to 24 μg/cm² protease allergen were identified from miRTarBase using the *multiMiR* R package. Pathway enrichment was then performed using Reactome pathways on target genes (n=3601) from upregulated DE miRNAs (n=13) and target genes (n=2746) from downregulated DE miRNAs (n=14) using the *enrichPathway()* function from the ReactomePA R package. The top 30 pathways according to highest gene ratio (were visualised using a dot plot generated in RStudio. The name of the enriched pathway is shown on the y-axes and the gene ratio (number of genes identified in the pathway / total genes used in the enrichment test) is indicated on the x-axes. Each dot represents an enriched pathway where the colour represents the significance of the association, and the size represents the number of genes involved in the pathway. (A) 30 out of the 164 significantly enriched pathways (Padj < 0.02) associated with upregulated miRNAs. (B) 30 out of the 464 significantly enriched pathways (Padj < 0.02) associated with downregulated miRNAs. The significance and gene count scales differ between plot (A) and plot (B).

Target gene-level enrichment was performed using experimentally validated target genes obtained from miRTarBase and assessed against Reactome pathways. For upregulated miRNAs (*n*=13), 164 pathways were significantly enriched (P-adjusted < 0.02) based on 3601 target genes (Supplementary Table 3). These included pathways associated with inflammatory signalling (IL-4, IL-13, IL-17, TNF, TLR and TGF-β signalling) and epithelial barrier regulation (WNT and NOTCH signalling) (Figure 6A; Supplementary Table 3). For downregulated miRNAs (n=14), 464 pathways were significantly enriched (P-adjusted < 0.02) based on 2746 target genes (Supplementary Table 4). This included pathways associated with inflammatory signalling (IL-4, IL-6, IL-7, IL-12, IL-13, IL-17, TLR and TGF-β signalling), immune responses (Fc receptor signalling), inflammatory cell death (pyroptosis), and epithelial barrier regulation (Rho GTPase signalling) (Figure 6B; Supplementary Table 4).

miRNA-level enrichment was performed using miEAA 2.0 against Reactome pathways from miRPathDB. For ORA, all miRNAs detected in the 24 μg/cm² protease experiment were used as the background universe (N=141). ORA identified two significantly enriched pathways (P-adjusted < 0.05) among upregulated miRNAs (n=13): Transcriptional activation of cell cycle inhibitor p2 and Transcriptional activation of p53 responsive genes. No significantly enriched pathways were identified among downregulated miRNAs (n=14) using ORA (P-adjusted < 0.05). miRNA-SEA was subsequently performed using all detected miRNAs ranked by logFC; however, no significant pathways were identified (P-adjusted < 0.05).

Investigating the biological function of DE miRNAs originating from Calu-3 cells

Pathway enrichment analysis was not performed for the DE miRNAs identified in Calu-3 cells because the small number of unique DE miRNAs (N=2) was expected to provide insufficient statistical power for enrichment analyses. Instead, a targeted literature search was conducted using PubMed to investigate the reported biological function of the identified miRNAs in airway biology, immune signalling, and allergic disease contexts. Searches were performed using each DE miRNA name (“miR-125b-5p”, “miR-574-5p”) in combination with the Boolean operator “AND” and the keywords “airway”, “immune”, “allergic”, “inflammation”, “epithelial”, and “extracellular vesicle”.

For miR-125b-5p, these searches returned 5 publications for airway, 85 for immune, 3 for allergic, 98 for inflammation, 44 for epithelial, and 89 for extracellular vesicle. Screening of these publications identified reports linking miR-125b-5p to inflammatory and immune-related processes, including macrophage regulation, as well as respiratory diseases including asthma and pulmonary fibrosis, and extracellular vesicle-mediated signalling (Supplementary Table 5).

For miR-574-5p, these searches returned 0 publications for airway, 12 for immune, 1 for allergic, 19 for inflammation, 13 for epithelial, and 20 for extracellular vesicle. Screening of these publications identified reports linking miR-574-5p to immune activation through Toll-like receptors (TLR) 7 and 8, regulation of epithelial–mesenchymal transition, and extracellular vesicle-mediated signalling (Supplementary Table 6).

Overall, substantially more publications were identified for miR-125b-5p than for miR-574-5p. Notably, miR-125b-5p was also the only differentially expressed miRNA observed in both the 24 µg/cm² protease-treated primary HBECs and the Can f 1-treated Calu-3 cells.

Discussion

This study established an EV-derived miRNA transcriptomic workflow using exclusively open-access software and applied it to investigate differential miRNA expression following allergen exposure in airway epithelial cells. Differential expression analysis identified biologically relevant EV miRNA signatures in the 24 µg/cm² protease-treated primary HBEC samples, with target gene-level enrichment implicating immune and inflammatory pathways, whereas interpretation of the Can f 1-treated Calu-3 dataset was constrained by low miRNA abundance and limited differential expression. Collectively, these findings demonstrate that EV-derived miRNA transcriptomic analysis can be performed without the use of commercial platforms, provided that appropriate experimental design supports robust downstream analyses, and highlight candidate miRNAs for further investigation.

The 24 µg/cm² protease-treated primary HBEC dataset provided a suitable model for developing and evaluating pathway enrichment approaches because it generated sufficient DE miRNAs (n = 27) to support biological interpretation. The increased miRNA abundance observed following high-dose protease exposure may reflect previous findings that EV production increases during lung inflammation, suggesting protease exposure may be triggering an inflammatory response (Holtzman and Lee, 2020). Target gene-level enrichment identified pathways associated with airway inflammation (IL-4 and IL-13 signalling) and allergic sensitisation (Fc receptor signalling) (Browne et al., 2025). These findings were further supported by the differential expression of miR-449a which has been implicated in IL-13-mediated epithelial differentiation in asthma (Solberg et al., 2012). Although miR-34b-3p was consistently differentially expressed across all protease exposure doses, limited evidence directly links this miRNA to airway inflammation; however, the miR-34 family has been implicated in airway biology and asthma and shares regulatory links with the miR-449 family (Solberg et al., 2012). Interestingly, the high-dose protease samples also exhibited increased rRNA and tRNA proportions, suggesting broader alterations in EV RNA cargo following protease exposure; however, further investigation is required to determine whether these changes represent biological responses or technical artefacts.

Biological interpretation of the Calu-3 dataset was constrained by the limited differential expression observed, likely reflecting low miRNA abundance and reduced statistical power, which precluded meaningful pathway enrichment analysis. Despite these limitations, literature-based analysis indicated that the Can f 1-associated DE miRNAs have previously been linked to immune and respiratory processes. miR-125b-5p has been implicated in inflammatory signalling, immune regulation, and respiratory diseases, suggesting that it may contribute to allergen-induced epithelial responses. miR-125b-5p was also detected in the 24 µg/cm² protease-treated primary HBEC samples, suggesting that this miRNA may represent a shared response across different allergen exposures and epithelial cell types. Fewer studies have investigated miR-574-5p in airway biology; however, it has recently been linked to innate immune activation through TLR7/8 signalling, supporting its potential involvement in allergen-induced epithelial immune responses. Overall, findings from the Calu-3 dataset should be interpreted cautiously due to reduced statistical power associated with insufficient miRNA abundance, and further work is required to validate these candidate miRNAs.

The discrepancy in significantly enriched pathways between target gene-level enrichment (n=628) compared to miRNA-level enrichment (n=2) may reflect differences in the level of annotation and database coverage used in each approach. Target gene databases can be extensive as miRNAs can target many genes and these have been extensively characterised, whereas miRNA-level databases are less comprehensive and therefore analysis is more restricted. Furthermore, pathway associations may have been inflated by the absence of context-specific miRNA target filtering, which was difficult to integrate into the workflow using open-source resources. This may have reduced the biological specificity of enriched pathways by retaining target genes not expressed in airway epithelial cells, consequently increasing the likelihood of false-positive pathway associations (Nazarov and Kreis, 2021). Additionally, this study focused on Reactome pathways, however alternative databases (e.g., KEGG and GO) may have provided additional miRNA-level associations and could be considered in future studies.

A major strength of this study was the development of the workflow using primary HBEC samples allowing optimisation using a dataset with sufficient differential expression. Additionally, the integration of RNA composition profiling and replicate assessment reduced the risk of technical variation being interpreted as biological signal. The use of experimentally validated miRNA-target interactions from miRTarBase improved confidence in pathway enrichment analysis, while the conservative interpretation of Calu-3 findings avoided over-interpretation of an underpowered dataset. Overall, the workflow enabled experimental limitations to be discovered which can subsequently be addressed to optimise further iterations of this work.

Several limitations should be considered when interpreting these findings. Differential expression analysis was constrained by the small number of biological replicates, reducing statistical power and restricting the removal of samples showing distinct PCA separation or variability between replicates. This allowed individual samples to disproportionately influence differential expression results (Liu et al., 2014). Interpretation of the Calu-3 dataset was further limited due to the low proportion of miRNA-assigned reads, reducing confidence that the observed differential expression reflected biological responses to allergen exposure rather than technical artefacts. The low miRNA yield may reflect the extensive dilution of EV preparations required to support EV quantification, cytokine analysis, and transcriptomics from the same samples (Browne et al., 2026). Substituting pathway enrichment analysis for a targeted literature review of Calu-3 derived DE miRNAs enabled biological interpretation but precluded the identification of novel pathway associations. Finally, pathway enrichment provides predictive rather than causal evidence and remains dependent on the completeness of functional annotations and biases towards pathways associated with well-characterised miRNAs, therefore the pathway-associations identified are not exhaustive (Godard and Eyll, 2015).

The limitations identified during this study provide several recommendations for future applications of the workflow. Future experiments should consider increasing the number of biological replicates and the amount of EV-derived RNA available for sequencing to enable robust differential expression analyses, particularly in Calu-3 experiments. Generating matched mRNA expression data using RNA-seq would improve the biological specificity of pathway enrichment analyses. Transcriptomic analysis could likewise be optimised. Network-based approaches such as miRNet to prioritise highly connected regulatory genes and alternative miRNA-set ranking strategies that combine effect size and significance could further enhance pathway enrichment analysis (Garcia-Moreno and Saez, 2020; Chang et al., 2020). Additionally, facilitating spike-in control data processing would negate technical variation and is especially useful for handling low-abundance miRNAs such as those identified in this study. The workflow should also be validated using established datasets and by comparing findings with the previously performed analyses (Hopkins et al., 2026; Browne et al., 2026).

Finally, the biological role of allergen-induced EV miRNA cargo could be further investigated by integrating matched mRNA and proteomic data to establish miRNA-mRNA-protein relationships. Candidate miRNAs should then be validated experimentally using RT-qPCR before functional studies employing miRNA mimics and inhibitors to determine their regulatory roles in immune and inflammatory pathways relevant to airway biology.

Overall, this study demonstrates that EV-derived miRNA transcriptomic analysis can be performed using exclusively open-source and open-access software, although integrating context-specific target gene filtering remains challenging. The workflow identified distinct RNA composition profiles and miRNA signatures in the 24 μg/cm^2 protease-treated primary HBEC samples, enabling comparison of pathway enrichment approaches. Target gene-level enrichment was found to provide the most informative biological interpretation for this dataset, identifying pathways related to immune signalling and epithelial barrier homeostasis.

Although pathway enrichment was unsuitable for the Calu-3 dataset because of limited differential expression, literature indicated that miR-125b-5p and miR-574-5p have previously been implicated in inflammatory signalling and respiratory disease. Experimental recommendations were provided to improve transcriptomic analysis in future allergen studies. Collectively, these findings demonstrate that biologically plausible EV-derived miRNA transcriptomic analysis can be achieved without commercial services, provided that experimental design supports robust downstream analyses.

Data Availability

Data generated for this analysis is available at the discretion of the corresponding authors: Browne et al., 2026 for Calu-3 data; Hopkins et al., 2026 for primary HBEC data. Test data has been uploaded to the GitHub corresponding to this study to demonstrate the workflow is functional. Test data consists of FASTQ files containing a subset of 1000 pseudo-random reads obtained from Khamina et al., 2022, available at the Gene Expression Omnibus (study accession: GSE189930, sample accession: SAMN23550840).

Acknowledgements

We thank Dr Georgina Hopkins and Dr William Browne for the contribution of the raw data. Particular thanks to Dr Georgina Hopkins, Dr Hannah Brown and Professor Lucy Fairclough for their support and guidance throughout the duration of this project. We acknowledge the use of Ada High-Performance Computing facility provided by the University of Nottingham which enabled the computation work.

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Supplementary Material

Supplementary material can be found appended to this document or, in the case of large tables and reports, on GitHub, accessible at: https://github.com/CJ770411/L4137_HBEC_EV_Transcriptomics.

Supplementary Table 1

Method	Description	Advantage	Disadvantage	Reference(s)
Experimental validation	Filter target genes according to the type and strength of experimental evidence supporting each miRNA-target interaction.	Can retain only target genes supported by stronger experimental evidence e.g., reporter assay, western blot and qPCR. Can exclude target genes supported by weaker experimental evidence e.g., microarray, NGS, pSILAC, CLIP-Seq, CLASH	Different experimental methods capture different aspects of miRNA regulation therefore filtering may bias towards well-studied genes or pathways. High-confidence support is not always available for true miRNA-target interactions. This can produce miRNA-target lists that are too small to support robust pathway enrichment analysis.	Huang et al., 2025; Cui et al., 2024
miRNA multi-targeting	Filter target genes according to the number of differentially expressed miRNAs known to target each gene, retaining genes co-targeted by multiple miRNAs. co-targeting a given target.	Prioritises genes that may be under stronger regulation by multiple miRNAs. Can improve the biological relevance of pathway enrichment by focusing on potentially highly connected regulatory genes.	Risks excluding real target genes that are only supported by one or a small number of miRNAs. The minimum threshold for gene co-targeting is arbitrary. Can bias towards better characterised genes which could influence pathway enrichment analysis.	Zhang et al., 2019; Hu et al., 2012
Multiple databases	Filter target genes according to the number of databases that report the miRNA-target interaction, retaining only interactions supported by multiple databases.	The intersection of 2 or more databases may reduce the likelihood of identifying false-positive target genes.	Due to differences in database sizes, curation methods and update frequency, real target genes present in only one, or a small number of databases may be excluded. The minimum threshold of databases required to support a miRNA-target interaction is arbitrary. Many databases are curated using the same data sources therefore consensus may not reflect independent support. Filtering by supporting databases may bias towards well-characterised miRNAs and target genes.	Hu et al., 2014; Xu et al., 2024; Huang et al., 2025
Context-specific filtering	Filter target genes according to whether expression in the biological system is plausible. Retain only the intersection of target genes identified and genes biologically plausible.	Excludes genes that are not relevant to the biological model (e.g., genes not expressed in airway epithelium), which may improve the specificity of pathway enrichment analysis by reducing noise and false-positive pathway associations.	Cells can express large proportions of the genome (e.g., respiratory ciliated cells can express ~12k genes), therefore this approach may not substantially reduce the target gene list. Airway epithelial-derived EVs can target multiple recipient cell types (e.g., immune cells) therefore it is difficult to define a suitable list of potential recipient cells. Creating an exhaustive list of cell types potentially targeted by airway-derived EVs would be very extensive and would likely incorporate a large proportion of the genome, limiting the effectiveness of this approach.	Bradley and Moxon, 2020

Network-based	Construct biological interaction networks to identify highly connected genes (hub genes). Prioritise genes with greater connectivity for downstream pathway enrichment analysis, as these genes may represent important regulators within the biological system.	Prioritising highly connected genes can reduce noise by excluding isolated target genes which may increase the specificity of pathway enrichment. Network-based approaches can incorporate additional biological information beyond individual miRNA-target predictions.	Defining connectivity thresholds is arbitrary and may exclude real miRNA-target interactions with less-connected roles. Network analyses depend on known connections, and this can bias towards better characterised genes.	Fan et al., 2016

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Supplementary Table 5

miRNA	Biological process / function	Disease / context	Search Keyword	Summary of reported findings	Reference(s)
miR-125b-5p	Macrophage regulation	Macrophages	Inflammatory	Regulates inflammatory state of macrophages through targeting of B7-H4.	Diao et al., 2017
miR-125b-5p	Airway inflammation	Asthma	Immune, allergic	Indicated in inflammatory response pathways and cilium assembly and organisation. Modulates macrophage polarization.	Roffel et al., 2022; Zhu et al., 2023
125b-5p	Potential biomarker	Cigarette smoke exposure	Airway	miR-125b-5p expression altered in gingival tissue following exposure to cigarette smoke	Sewer et al., 2020
miR-125b-5p	Pulmonary fibrosis	Pulmonary fibrosis in mice	Epithelial	miR-125b-5p interacts with BAK1 to regulate lung fibrosis and the epithelial-mesenchymal transition in lung epithelial cells.	Zhou et al., 2024

miR-125b-5p	Extracellular vesicle signalling	Extracellular vesicles	Extracellular vesicle, inflammation	Small extracellular vesicles transfer miR-125b-5p to recipient pulmonary cells, reducing inflammation through regulation of the TRAF6 signalling pathway. Adipose-derived stem cell exosomes deliver miR-125b-5p to pulmonary endothelial cells, reducing ferroptosis via the Keap1/Nrf2/GPX4 pathway.	Peng et al., 2023; Shen et al., 2023

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Supplementary Table 6

miRNA	Biological process / function	Disease / context	Search Keyword	Summary of reported findings	Reference(s)
miR-574-5p	Immune regulation	Toll-like receptor	Immune, inflammation	miR-574-5p regulates TLR8/7, with signalling abnormalities associated with autoimmune diseases (e.g., lupus) and inflammation-related cancers, and neonatal respiratory distress syndrome via interactions with RAB10.	Yang, 2024; Wang et al., 2024; Ji et al., 2026

miR-574-5p	Regulation of epithelial-mesenchymal transition	Small cell lung cancer	Epithelial	miR-574-5p binds directly to vimentin to potentially mediate the epithelial-mesenchymal transition in small cell lung cancer.	Sun et al., 2021
miR-574-5p	Extracellular vesicle signalling	Extracellular vesicles	Extracellular vesicle	Small extracellular vesicles transfer miR-574-5p to activate TLR7/8 and regulate prostaglandin E_2 in adenocarcinoma.	Donzelli et al., 2021

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